ALKENE

161. (d) Solution of bromine in carbon tetrachloride is used to test for unsaturation of alkene. Red colour of bromine disappears due to the formation of colourless dibromo ethane $(C_2H_4Br_2)$.

162. (b) $\underset{\text{Propylene}}{CH_2=CH-CH_3} + H_2O \xrightarrow{H_2So_4} \underset{\text{Isopropyl alcohol}}{CH_3-\overset{\overset{OH}{|}}{C}H-CH_3}$

Thus in this reaction isopropyl alcohol is formed.

163. (b) Propene gives formaldehyde and acetaldehyde on ozonolysis.

$$CH_2=CH.CH_3 \xrightarrow{O_3} \overset{\overset{O}{\diagup\diagdown}}{\underset{\underset{O}{|}}{CH_2}-\underset{\underset{O}{|}}{CH}.CH_3} - \xrightarrow[H_2O_2]{Zn/H_2O} HCHO + CH_3CHO$$

164. (a) When ethylene is treated with cold alkaline $KMnO_4$, ethylene glycol is formed.

$$\begin{matrix}H\\H\end{matrix}\!\!>C=C<\!\!\begin{matrix}H\\H\end{matrix} \xrightarrow[\text{Cold alkaline}]{KMnO_4} \begin{matrix}H\\H\end{matrix}\!\!>\underset{\underset{OH}{|}}{C}-\underset{\underset{OH}{|}}{C}<\!\!\begin{matrix}H\\H\end{matrix}$$

Ethylene glycol

165. (a) $C_6H_6 + H_2C=CH_2 \xrightarrow[95^\circ \text{ pressure}]{AlCl_3.HI} C_6H_5CH_2CH_3$

166. (a) Markownikoff as well as anti-Markownikoff's rule is valid only for unsymmetrical alkenes.

167. (b) The formation of *n*-propyl bromide in presence of peroxide can be explained as follows.

Step-1: Peroxide undergo fission to give free radicals $R-O-O-R \rightarrow 2-R-\dot{O}$

Step-2 : HBr combines with free radical to form bromine free radical $R-\dot{O}+HBr \rightarrow R-OH+B\dot{r}$

Step-3 : $B\dot{r}$ attacks the double bond of the alkene to form a more stable free radical

$$CH_3CH = CH_2 + B\dot{r} \rightarrow \begin{cases} CH_3 - \dot{C}H - CH_2Br \text{ (more stable)} \\ CH_3 - \overset{Br}{\overset{|}{C}}H - \dot{C}H_2 \text{ (less stable)} \end{cases}$$

Step-4 : More stable free radical attacks the HBr

$$CH_3 - \dot{C}H - CH_2 - Br + HBr \rightarrow \underset{\text{n-propyl bromide}}{CH_3CH_2CH_2Br} + B\dot{r}$$

Step-5 : $B\dot{r} + B\dot{r} \rightarrow Br_2$

168. (c) $CH_2 = CH_2 \xrightarrow{KMnO_4} \underset{OH}{\underset{|}{C}H_2} - \underset{OH}{\underset{|}{C}H_2}$

$CH_2 = CH_2 \xrightarrow{AgNO_3}$ No reaction

169. (b) According to Markownikoff's rule, the negative part of the unsymmetrical reagent adds to less hydrogenated (more substituted) carbon atom of the double bond.

$$C_6H_5CH = CH - CH_3 + HBr \rightarrow C_6H_5\overset{Br}{\overset{|}{C}}HCH_2CH_3$$

171. (a)

$$\underset{CH_2}{\underset{\|}{CH_2}} + H_2SO_4 \rightarrow \underset{CH_2HSO_4}{\underset{|}{CH_3}} \xrightarrow{H_2O} \underset{CH_2OH}{\underset{|}{CH_3}} + H_2SO_4$$

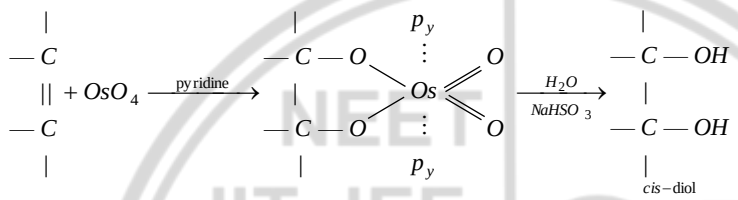
Except ethyl alcohol, no other primary alcohol can be prepared by this method as the addition of H_2SO_4 follows Markownikoff's rule. Generally secondary and tertiary alcohols are obtained.

172. (c) $CH_3 - \underset{CH_3}{\underset{|}{C}} = CH_2 \xrightarrow[\text{anti. Markownikoff's addtion}]{HBr\ /\ peroxide} CH_3 - \underset{CH_3}{\underset{|}{C}}HCH_2 - Br$

173. (c) Decolourisation of $KMnO_4$

Ethene	Acetylene
Cu_2Cl_2, NH_4OH	Cu_2Cl_2, NH_4OH
no. ppt.	red ppt.

174. (a)



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